
accelerate
collaborate
innovate
communicate
palm
PalmSource
Conference 2000

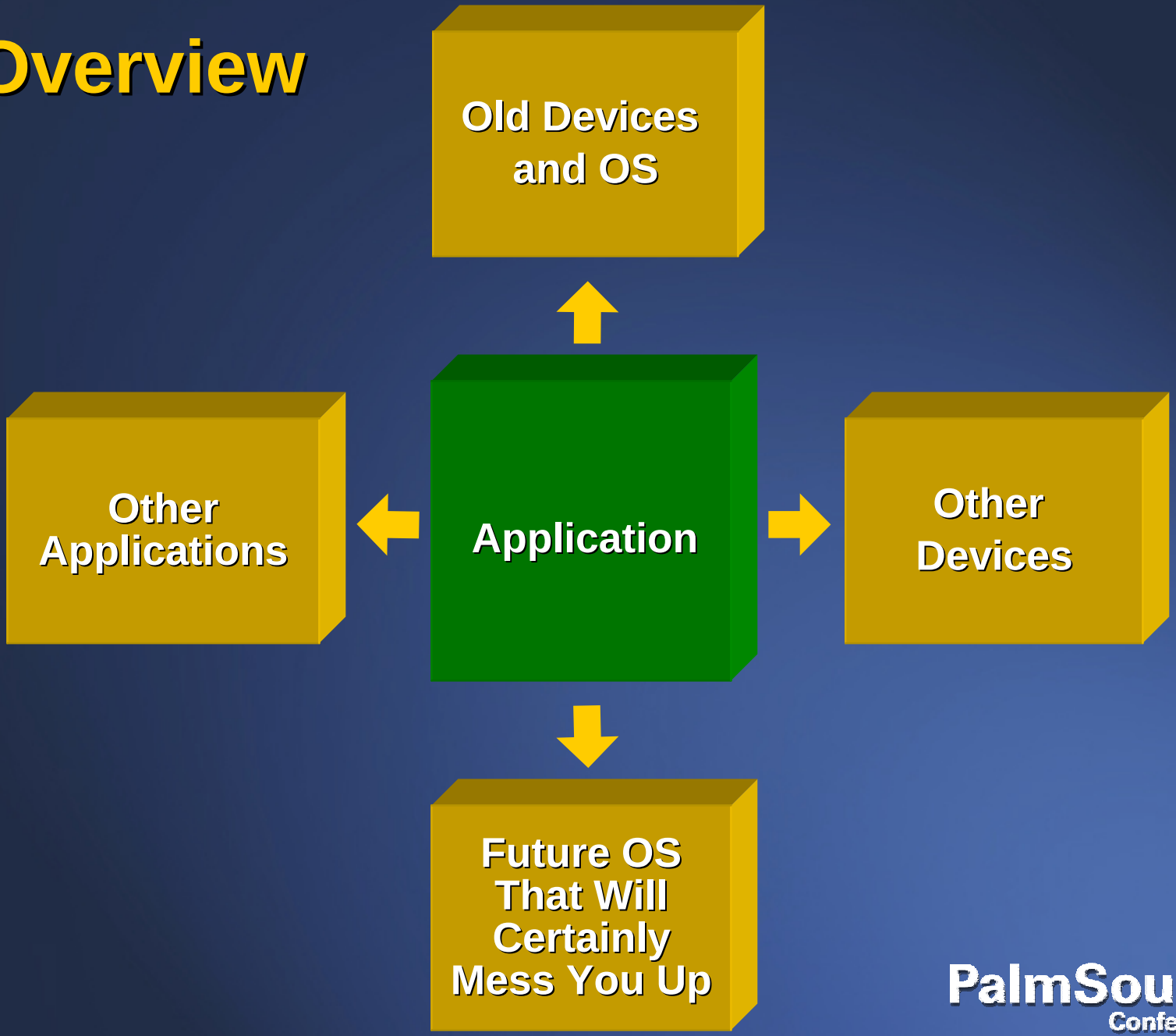


101: OS and Device Cross Compatibility

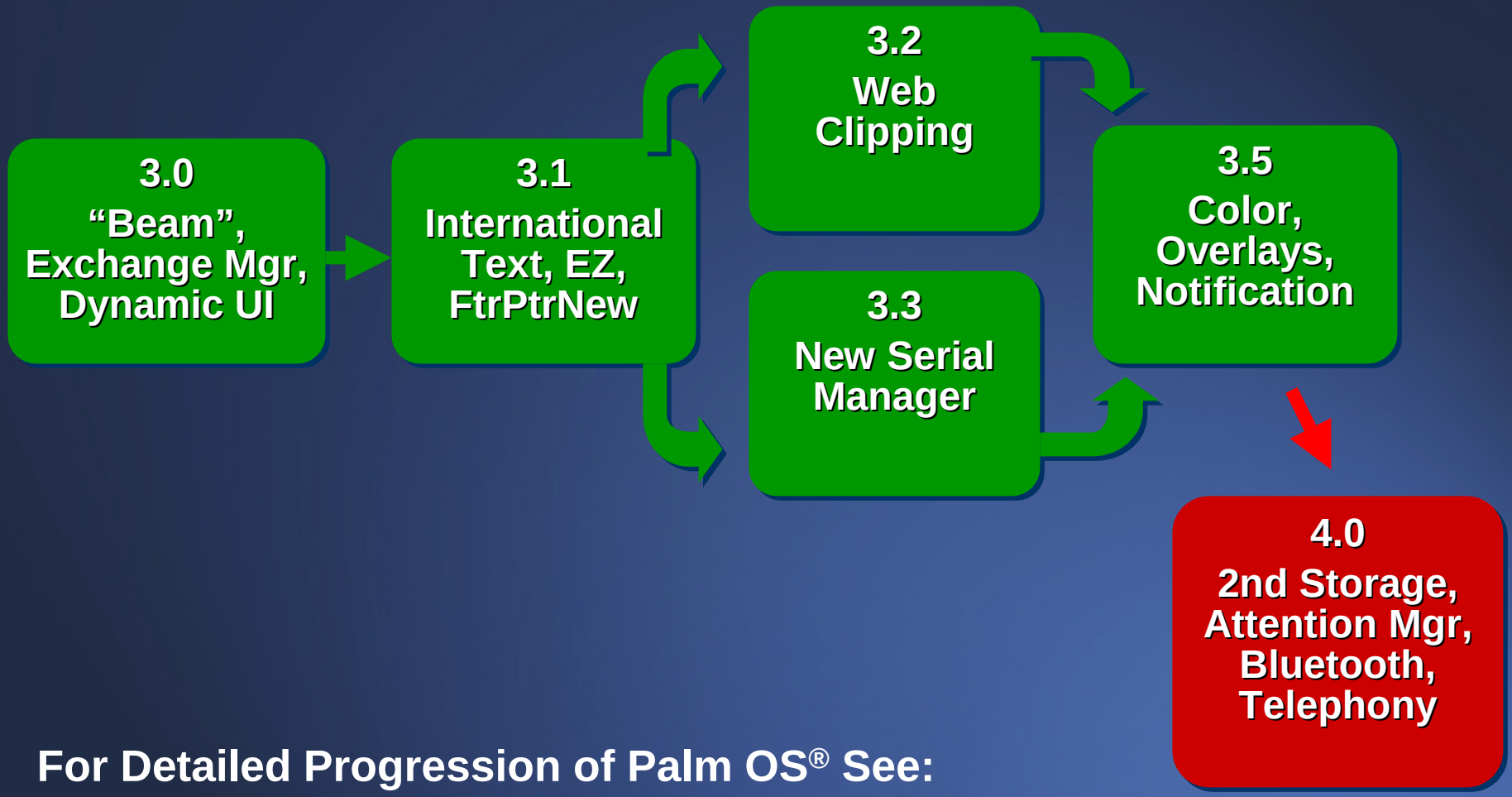
Ezekiel Sanborn de Asis
DTS Engineer

PalmSource
Conference 2000

Overview


Old Devices and OS
Other Applications
Application
Other Devices
Future OS That Will Certainly Mess You Up

The Road to Here



For Detailed Progression of Palm OS® See:

Palm OS Reference -> Libraries -> Compatibility Guide

PalmSource Conference 2000

Backwards Compatibility

- Program with the most up-to-date SDK
- Decide on a minimum feature set
- Check for
  - Minimum OS version
  - The existence of components that you are going to use
- Some features are introduced in between OS versions and may appear on earlier devices
  - VFS Manager
  - Direct Color
  - Generally, abide by the docs

Checking for Components

```
Err err = FtrGet(sysFtrCreator, sysFtr..., &value);
if (value && (err!=ftrErrNoSuchFeature))
   //… is there
```

- sysFtrCreator is the creator ID of app that “owns” this feature
  - For this purpose you will often, but not always, use sysFtrCreator
- sysFtr… is a place holder for the feature you are checking on
  - Look at SystemMgr.h for a complete list
- ROM version check is good example

Dynamic UI—3.0

- If you need to use it you will have to set the minimum ROM to 3.0
- Dynamic UI was always a little hairy…
  - This has been fixed in 4.0
  - For earlier OS versions consider, adding the UI elements to the form and hiding and showing them as needed
  - You can change the text of, move and resize most UI elements (exception being tables) with standard APIs

Backwards Compatibility

Beaming—3.0

- Either turn off Beaming functionality for older devices or restrict to 3.0 and up

File Streaming—3.0

- Nothing to do but restrict to 3.0 and up

Dragonball EZ CPU—3.1

- As long as you are not accessing the CPU directly you are okay. If you are
  - Check the processor (see compatibility guide)
  - VZ and ARM possible in 3.5 and higher

International Feature Set—3.1

- New functions were added to 3.1 and 3.1J to support the possibility of 2 byte characters
- Use PalmOSGlue.lib functions
  - It uses version in OS if exists
  - Text Manager search, compare, manipulate text, and get character attributes
  - TxtGlueGetNumericSpaceChar and TxtGlueGetHorizEllipseChar to use numeric space and horizontal Ellipse
  - TxtGlueCharIsVirtual
  - Lower and Upper case transliteration
- Use sysFtrNumEncoding in previous code bit to get character encoding

Web Clipping—3.2

- Not in 3.3 and in some 3.5 devices…
  So check for clipper instead of OS version

```
err = DmGetNextDatabaseByTypeCreator(true,
  &searchState, sysFileTApplication,
  sysFileCClipper, true, &cardNo, &dbID);
if (!err)
   //clipper is there
```

OR check for the Inet.lib

```
err = SysLibFind("INETLIB", pRefNum);
if (!err)
  //inet.lib is not loaded
```

New Serial Manager—3.3

- Old serial calls still work for now
- For Virtual Drivers or Connection Manager check for new serial manager

```
err = FtrGet(sysFileCSerialMgr,
  sysFtrNewSerialPresent, &value);
```

Color—3.5

- Determine bit depth of device
  - SDK ≥ 3.5 : WinScreenMode to get bit depth
  - SDK < 3.5 : ScrDisplayMode to get bit depth
- 1 App or 2
  - You can write your app to act appropriately based on the max bit depth of the device
  - Color Apps may be large for a B/W device with less storage memory… So you may want to create separate apps
- Use Bitmap Families and make sure that the 1st member of the family is a 1-bit B/W image

Graphical UI—3.5

- As long as the usable bit is not set they will remain hidden and not cause a problem on old versions of the OS
- Consider creating a normal (bitmap covered if you like) button and a graphical button and then turn the usable bit on the appropriate one based on the OS version

```
If (OSVersion >= 350S){
      FrmShowObject(pForm, myGraphButton);
      FrmHideObject(pForm, myStandardButton);
}
```

New Gadgets—3.5

- You can use old style gadgets
- Avoid overlapping gadget and other UI element
- Use 3.5 Gadgets for 3.5 and up apps

Other 3.5 Issues

- Use the WinDraw… function instead of WinPaint… unless you know you're in 3.5 or above
- Do not draw before FrmDrawForm is called
- There is no way to use overlay pre 3.5
- Security
  - prefShowPrivateRecords on 3.5 OS
  - prefHidePrivateRecordsV33 on older OS
- Clear out event structure before adding to queue
- Check for Notification Manager before using

Future Compatibility

What might change on you

- New Hardware
- Move the physical location of buttons, etc.
- Change the processor
- Change the screen size (not 160 by 160)
- Change Formats of PIM

Moving to Palm OS® 4.0

- There should not be many incompatibilities moving to OS 4.0
  - Start checking with 4.0 debug ROMs
  - Improved checking in 4.0 Debug ROMs
- Applications may run off of expansion card
  - Do they require the presence of something else in the system?

5.0 and Beyond

- In case you have not heard



How to Make App Run on ARM

- Use only documented APIs as they are intended
- Do not patch traps
- Never open up a system's struct
  - In 3.5 sometimes the API is lacking
  - There is no good API to look in BitmapType Struct
  - In 4.0 this is fixed
  - With 4.0 you can disallow system struct access with

```
#define DO_NOT_ALLOW_ACCESS_TO_INTERNALS_OF_STRUCTS
```

- Test with 3.5 or 4.0 Debug ROM (all checks turned on) and fix anything it complains about

Screen Access

- Do not write directly to the screen
  - Build your image in an offscreen buffer located in RAM
  - Copy the whole thing with WinCopyRectangle
- If you NEED the extra bit of speed, let the user switch between an optimized and legal version

Device Compatibility

- Greater variety of devices with Palm OS®
- Hardware differences
  - Bar scanner, jog dial, springboard, etc.
  - Screen size
  - Physical shape
- Early forms of APIs from next OS

Symbol With OS 3.5

- Unique Hardware
  - Bar Scanner
  - Wireless Ethernet
- Unique layout of buttons
  - Read hard keys via the system



+ Wireless Ethernet

PalmSource Conference 2000

Clie With OS 3.5

- Unique Hardware
  - Jog Dial
  - Can write code to handle events because you should not get them if not on a Sony
- Memory Stick
  - Same VFS API as in Palm OS® 4.0
  - If you use open APIs from Sony it should still work on any OS 4.0 device
  - FtrGet to find if the expansion manager is there



Handspring With OS 3.5

- Unique Hardware
  - Springboard
- VZ processor
  - You should not be playing directly with the processor anyway
- 16-bit Color APIs the same as you will find on OS 4.0



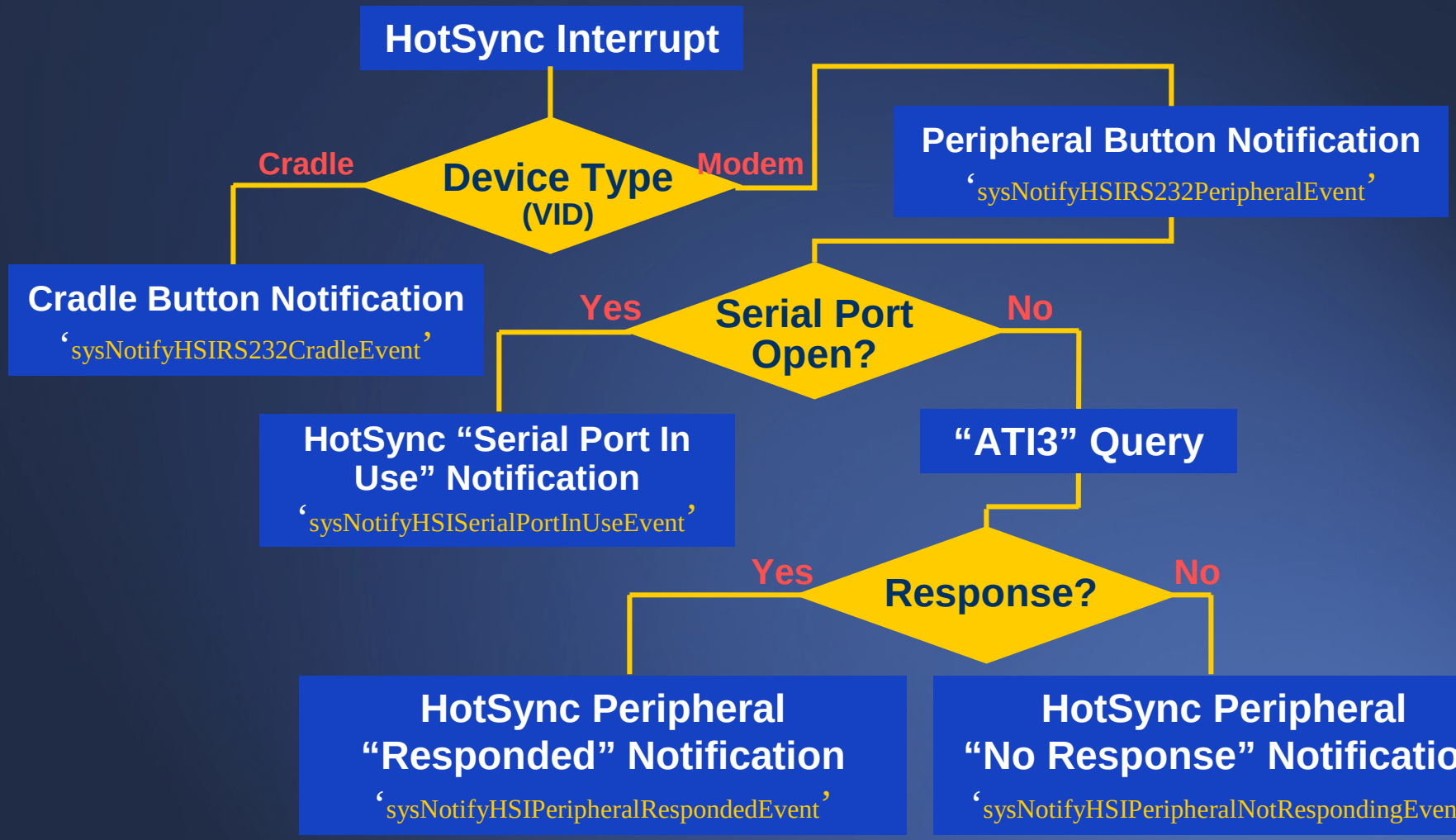
Checking for 16-bit Color

```
  // See if OS supports color or not
FtrGet (sysFtrCreator, sysFtrNumROMVersion, &version);
if (version > 0x03500000)
{
  void* procP;
  AppOSSupportsColor = true;
  // See if it supports 16-bit color
  procP = SysGetTrapAddress(sysTrapWinSetForeColorRGB);
  if (procP && procP!= SysGetTrapAddress
      (sysTrapSysUnimplemented))
    AppOSSupportsDirectColor = true;
  else
    AppOSSupportsDirectColor = false;
}
```

Peaceful Coexistence

- You application should never interfere with another application
  - As always do not step on the system
  - Avoid patching traps
  - Return the same system you got (fonts, draw state)

Using the HotSync® Interrupt Mgr



Allow for the Attention Manager

- Subtle and Insistent method to get users attention that something has occurred
- Subtle method appears as a flashing icon in title bar
- Other apps depend on this working
- DO NOT inhibit Insistent method
- If you do not have a title bar, there must be a logical method to relay the subtle attention


accelerate
collaborate
innovate
communicate
palm
PalmSource
Conference 2000